EXERCISE 2: SQL AGGREGATES & OPERATORS

1. SELECT DISTINCT department
   FROM students_table;

department
IT
HR
Finance

2. SELECT AVG(age) AS avg_age, department
   FROM students_table
   GROUP BY department;

department	avg_age
IT	21.5
HR	21
Finance	22

3. SELECT COUNT(*) AS student_count, department
   FROM students_table
   GROUP BY department
   HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

4. SELECT student_id,
   name,
   age,
   department
   FROM students_table
   WHERE age BETWEEN 21 AND 23;

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StudentID	Name	Age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR.

5. SELECT student_id,
name,
age,
department
FROM students_table
WHERE department IN ('IT', 'HR')
AND age > 21;

StudentID	Name	Age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT SUM(credits) AS total_credits,
department
FROM courses_table
GROUP BY department
HAVING SUM (credits) > 5;

department	total_credits
IT	11

7. SELECT course_id,
course_name,
department,
credits
FROM courses_table

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7. WHERE credits != 4;

course_id	course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
credits
FROM courses_table
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT MAX(grade) AS max_grade,
MIN(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM enrollments_table;

max_grade	min_grade	avg_grade
90	78	84.6

$$\frac{(85+78+90+88+82)}{5} = 84.6$$

calculate avg_grade

10. SELECT COUNT(*) enrollment_count, course_id
FROM enrollments_table
GROUP BY course_id;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. SELECT SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus,
department
FROM ~~employees~~ Salaries_table
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12. SELECT AVG(Salary) AS avg_salary,
department
FROM Salaries_table
GROUP BY department
HAVING AVG(Salary) > 55 000;

department	avg_salary
IT	61 000
Finance	70 000

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13. SELECT employee_id,
name,
salary,
bonus,
(salary + bonus) AS total_compensation
FROM salaries_table
WHERE (salary + bonus) > 60 000;

employee_id	Name	Salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5500	67 500

14. SELECT SUM(budget) AS total_budget,
AVG(budget) AS avg_budget,
department
FROM projects_table
GROUP BY department
HAVING AVG(budget) > 70 000;

department	total_budget	avg_budget
IT	300 000	150 000
Finance	80 000	80 000

15. SELECT project_id,
project_name,
department,
budget
FROM projects_table
WHERE budget BETWEEN 50 000 AND 120 000
AND department != 'Marketing';

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projectid	projectname	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
5	HR Portal	HR	50 000

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